

SOMA Runtime: Complete Analysis & Strategy to Beat System 0

1. THE ENEMY: SYSTEM 0 (Pin-to-Pin)

What System 0 Actually Is (Helix 02, January 2026)

Architecture:

- 10M parameter neural network
- Input: full-body joint state + base motion (from S1 at 200Hz)
- Output: joint-level actuator commands at 1 kHz
- Replaces 109,504 lines of hand-engineered C++

Training:

- 1,000+ hours of human motion capture data
- Retargeted to robot's joint structure
- Sim-to-real RL across 200,000+ parallel environments
- Massive domain randomization for zero-shot real-world transfer
- Training cost: estimated 1000s of GPU-hours on large clusters

What it does:

- Whole-body locomotion + manipulation simultaneously
- Walk while carrying, adjust balance while reaching
- 4-minute autonomous dishwasher task (61 loco-manipulation actions)
- Contact management, gait generation, balance, impedance — ALL in one network

What it does NOT have:

- Safety certification (no CBF, no formal guarantees)
- Frequency invariance (fixed 1kHz, retrain for different rates)
- Robot-agnostic deployment (trained per robot morphology)
- Open architecture (black box, proprietary to Figure)

The Full Stack: S0 + S1 + S2

S2 (VLM, 7B params, 7-9 Hz): Language understanding, scene reasoning, goal setting

↓ latent semantic representation

S1 (Visuomotor, 80M params, 200 Hz): Vision → full-body joint targets

↓ joint position/velocity targets

S0 (Motor, 10M params, 1 kHz): Joint targets → safe motor torques



SOMA replaces S0. Not S1, not S2. The motor control layer.

2. THE COMPETITORS (What Everyone Uses)

Runtime A — Arms/Manipulators (PROVEN)

Robot	Controller	Rate	What SOMA Replaces
Franka Panda	Firmware PD + FCI torque API	1kHz	Dumb PD → intelligent CfC+CBF via libfranka
KUKA iiwa	Impedance/PD in KRC	1kHz	Proprietary firmware → portable SOMA
UR5/10	PID in control box	500Hz	Position PID → torque-level CfC
ABB IRB	Proprietary IRC5 PID	250Hz	Closed-box → open SOMA

Status: v7 PROVEN. 2.37° tracking, zero safety violations, 133μs, frequency invariant.

Runtime B — Legged/Humanoids (IN PROGRESS)

Robot	Controller	Rate	Training
Figure 03	System 0 (10M NN)	1kHz	1000hrs mocap + 200K env RL
Atlas (BD)	Custom MPC + RL	~200Hz	Years of PhD-level hand-tuning
Spot (BD)	Dozens of parallel MPCs + RL selector	<1ms	Proprietary
Unitree Go2	RL policy (PPO in IsaacGym/MJX)	500Hz	Teacher-student distillation
Unitree G1	RL velocity tracking	~500Hz	PPO + domain randomization

Status: v8 residual beats PD on Go2 push recovery. Proven at force boundaries.

3. WHAT WORKED (Hard Evidence)

v7 — Franka 7-DOF Arm (Runtime A)

Metric	Result	Method
Tracking accuracy	2.37° overall	Dagger (3 rounds) on computed-torque expert

Metric	Result	Method
Ramp tracking	1.51°	
Waypoint tracking	2.05°	
Step tracking	1.92°	
Inference latency	133μs (7,500 Hz)	CfC-NCP, 134K params, 527KB
Safety violations	ZERO under 15N perturbation	CBF-QP analytical filter
Frequency invariance	200Hz–2kHz, same weights	CfC dt-gate, 2.50° spread
C SDK	bit-for-bit match to PyTorch	Single-header, zero deps
Training time	~20 min DAgger	T4 GPU

What made it work:

1. **Inspect-first approach** — v7 started by checking the URDF, finding the Joint 4 range bug that killed v1-v6
2. **Clean expert data** — Once the actuator bug was fixed, iLQR/computed-torque gave consistent demonstrations
3. **DAgger** — 3 rounds closed the distribution shift that pure BC couldn't handle
4. **Analytical CBF** — Not learned, not neural. Exact barrier functions on joint limits, velocity limits, torque rates
5. **Variable-dt training** — Random timesteps 0.5ms-20ms → frequency invariance baked into training

v8 — Go2 Quadruped (Runtime B)

Metric	Result	Method
Impulse recovery	350N (PD: 300N, +17%)	Residual CfC on PD
Sustained push	120N/200ms (PD: 100N, +20%)	
Diagonal push	500N (PD: 400N, +25%)	
Tilt reduction	9-25% when both survive	Recurrent state advantage
Inference latency	65μs (15,315 Hz)	25,868 params, 101KB
Training time	21 min	custom_vjp through MJX
CBF latency	25μs (39,372 Hz)	Analytical

What made it work:

1. **Residual architecture** — $\text{ctrl} = \text{PD}(\text{state}) + 0.3 \cdot \text{CfC}(\text{state})$. CfC doesn't learn gravity comp (PD has it).
CfC only learns what PD can't: predictive push recovery
2. **custom_vjp** — Full-fidelity MJX forward pass, finite-difference backward through 12 control dims.
Solves MJX's reverse-mode AD failures
3. **Fixed horizon** — Single JIT compile for all phases. No recompilation overhead
4. **Force curriculum** — Progressive: 10→30, 20→60, 40→100, 60→150N
5. **CfC recurrent state** — Carries temporal information (push history, oscillation phase) that PD can't access

HyperFlux Gaming Kernels (Shipped Production)

Kernel	Speedup	Latency
Ballistics v15.5	867x	4.8μs
Hitbox v2.2	172x	Sub-15μs
10 kernels total	Various	C++ SDK delivered

4. WHAT FAILED (Critical Lessons)

The 13-Version Bug (v1-v5.0)

Root cause: Joint 4 starting position was outside its valid range $[-3.07, -0.07]$ in every version from v1 to v5.0. This caused OSQP to go 80% infeasible, which was misdiagnosed as "MPC doesn't work" and led to:

- v3.x: Blamed OSQP, tried DAgger on bad data
- v4.x: Blamed distribution shift, tried Simplex mode-switching
- v5.0: Panicked, switched to PPO (wrong paradigm for CfC)

Lesson: ALWAYS inspect the model first. Bad data poisoned 13 versions.

Replay vs Closed-Loop (v4.1)

- Replay evaluation: 100% PERFORMANCE mode
- Closed-loop evaluation: 0.2% PERFORMANCE, 99.4% FALLBACK

Lesson: Never trust replay metrics. Always evaluate closed-loop from day 1.

Raw Torque CfC (v8 first attempt)

CfC outputting raw torques fell at 0N (every force level, including baseline). Loss converged (196→1.95) but in MJX with patched geometry. Eval in native MuJoCo with full geometry = instant collapse.

Root cause: Asking a 25K network to learn gravity compensation + balance + push recovery from scratch in 80 gradient steps is impossible when PD has analytical gravity comp.

Lesson: Residual architecture is mandatory for fast training. Start from working controller, learn only the delta.

Dagger Contamination (v7 R2-R3)

Training loss INCREASED from R1 to R3 (0.00463 → 0.01733) as the DAgger pool grew. More data from badly diverged states contaminated the expert distribution.

Lesson: Stratify DAgger. Oversample underperforming trajectories, undersample solved ones. Weight expert data 2x.

MJX Differentiability Failures (v8 early attempts)

MJX's Newton solver uses `lax.while_loop` which JAX cannot reverse-mode differentiate through. Direct `jax.grad(loss)` through `mjx.step` crashes.

Lesson: custom_vjp is the correct approach. Full-fidelity forward, finite-difference backward through control dims only. 24 extra mjx.step per physics step (12 dims × 2 for central FD).

PPO Pivot (v5.0-v5.1)

PPO destroys the analytical properties (Lyapunov stability, frequency invariance) that are SOMA's moat. The reward function creates a black-box policy.

Lesson: DiffSim + DAgger preserve CfC structure. PPO/SAC are last resort.

Sinusoid Tracking (v7)

Sinusoid stayed at 56-66° across all DAgger rounds while ramp/waypoint/step converged. High-frequency tracking at 50° amplitude with torque-limited wrist joints was physically untrackable.

Lesson: Not all tracking targets are realistic. 3/4 trajectory types were production-ready. Sinusoid was a diagnostic, not a use case.

5. THE DATA QUESTION

What System 0 Uses (and why)

1,000 hours of human motion capture data:

- Humans wearing mocap suits performing diverse tasks
- Retargeted to Figure 03's joint structure

- Provides natural movement priors: gait timing, arm swing, balance strategies
- This is the "style" teacher — makes the robot move like a human

200,000 parallel simulation environments:

- Each with different physics parameters (friction, mass, damping)
- Domain randomization for sim-to-real transfer
- PPO learns a policy robust to parameter uncertainty

Why this works for them: They have ONE robot (Figure 03). They can spend months tuning. They have \$2.6B in funding and massive GPU clusters.

What SOMA Uses (and why it's different)

SOMA v7 data (proven on arms):

- URDF file (robot description) → analytical expert (iLQR/computed-torque)
- BC training: ~150K expert state-action pairs
- DAgger: 3 rounds adding ~60K correction samples each
- Total: ~330K samples, generated automatically

SOMA v8 data (proven on quadrupeds):

- URDF file → MJX differentiable simulation
- NO expert data. NO demonstrations. NO mocap.
- DiffSim: direct gradient through physics
- 160 training steps × 4 batch × 50-step rollouts = 32,000 physics rollouts
- 21 minutes on one T4 GPU

The fundamental difference:

	System 0	SOMA
Human data	1,000 hours mocap	ZERO
Expert demonstrations	Teleoperation	Analytical (auto-generated from URDF)
Sim environments	200,000 parallel	4-8 batch (custom_vjp)
Training time	Weeks on GPU cluster	21 minutes on T4
Per-robot effort	Months of tuning	20 min from URDF
New robot	Start over	Swap URDF, retrain

What Data We Actually Need for Full Runtime

For standing + push recovery (PROVEN):

- Just the URDF. DiffSim generates everything.

For trajectory tracking (PROVEN on arms):

- URDF + analytical expert (iLQR/computed-torque). Auto-generated.

For walking (NOT YET PROVEN):

- This is the hard one. Options:

Option A: Pure velocity tracking (no human data)

Input: velocity command (v_x, v_y, ω) + current state

Output: joint torques that achieve that velocity

Loss: $\| \text{actual_velocity} - \text{commanded_velocity} \|^2 + \text{balance} + \text{energy}$

- Train with DiffSim through long rollouts (500+ steps)
- CfC must discover gait patterns autonomously
- Risk: gradient explosion over long horizons, may converge to non-walking solution

Option B: Reference gait from optimization (minimal data)

1. Generate reference gaits via trajectory optimization (TOPP-RA, CasADi)

2. Use as loose reference in DiffSim cost: reward similarity to reference

3. CfC learns to execute + adapt the gait

- Reference generated analytically (no human data)
- CfC free to deviate when needed (push recovery, terrain)
- Moderate risk: requires good reference trajectory generator

Option C: RL teacher + CfC distillation (following Unitree's approach)

1. Train PPO teacher in IsaacGym/MJX (standard RL pipeline)

2. Collect teacher rollout data

3. Distill into CfC via BC + DAgger

4. CfC inherits RL behavior but with CfC structure (dt-gate, $O(1)$)

- Well-proven by Unitree (their entire Go2/G1 pipeline)
- Requires RL infrastructure but not human data
- Risk: PPO destroys CfC analytical properties → need careful distillation

Option D: Human motion reference (following Figure's approach)

1. Use public mocap datasets (CMU Mocap, AMASS, HumanML3D)
2. Retarget to robot morphology
3. Train CfC to track retargeted motions via DiffSim
4. Add domain randomization for robustness

- Uses existing public data (no need to collect our own)
- Produces natural-looking movement
- Risk: retargeting quality, large dataset processing

My Recommendation: Hybrid B+C

Phase 1 (NOW → 2 weeks): Option B — Reference gait + DiffSim

- Generate simple walking gait via trajectory optimization
- Train CfC residual to track it through DiffSim
- Proves: "SOMA walks, from URDF, in 20 minutes"

Phase 2 (2-4 weeks): Option C — RL teacher for complex behaviors

- Train PPO teacher for diverse locomotion (speed changes, turns, stairs)
- Distill into CfC with DAgger
- Proves: "SOMA handles diverse locomotion"

Phase 3 (1-2 months): Option D — Natural motion from public mocap

- Import AMASS/CMU motion clips
- Train CfC to reproduce natural human movement patterns
- Proves: "SOMA moves like a human, certified safe"

This sequence avoids the biggest risk (raw DiffSim walking from scratch) while building toward full System 0 capability.

6. HOW TO BEAT SYSTEM 0

Where System 0 is Unbeatable (don't compete here)

- **Natural movement quality** — 1,000 hours of human data gives movement "style" we can't match with 0 data
- **Emergent behaviors** — 200K parallel envs discover behaviors we can't program

- **Scale** — \$2.6B funding, hundreds of engineers, massive GPU clusters

Where SOMA Wins (our actual moat)

1. Speed to deploy: 20 minutes vs months

Customer gives URDF → SOMA trains CfC+CBF → delivers libsoma.so
↓
20 minutes on 1 GPU

System 0 requires months of per-robot tuning, mocap data collection, RL training on large clusters. For the 100+ humanoid companies that aren't Figure, they can't afford this.

2. Certified safety: CBF vs nothing

SOMA: CfC(state) → CBF filter → motors
↓ ↓
fast control PROVEN safe (joint limits, velocity limits,
torque rates, orientation — all certified)

System 0 has zero formal safety guarantees. ISO 13849, EU AI Act will require safety certification for commercial deployment. SOMA is the only architecture designed for this.

3. Size + Speed: 25K params / 15kHz vs 10M params / 1kHz

Metric	SOMA	System 0	Advantage
Parameters	25,868	10,000,000	386x smaller
Control rate	15,315 Hz	1,000 Hz	15x faster
Model size	101 KB	~40 MB	400x smaller
Compute	Single CPU core	Embedded GPU	No GPU needed

This matters for: embedded deployment (no GPU), battery life, reliability, cost.

4. Frequency invariance: any rate, same weights

Train once at random dt → deploy at 200Hz, 500Hz, 1kHz, 2kHz, 10kHz
Same weights. Same performance.

System 0 is trained at 1kHz. Changing the rate requires retraining. SOMA's CfC dt-gate ($\sigma = \text{sigmoid}(\tau \cdot dt + b)$) handles this structurally.

5. Robot-agnostic: same code, any URDF

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Franka → SOMA(franka.urdf) → libsoma_franka.so ← 20 min
Go2 → SOMA(go2.urdf) → libsoma_go2.so ← 20 min
G1 → SOMA(g1.urdf) → libsoma_g1.so ← 20 min
Custom → SOMA(custom.urdf) → libsoma_custom.so ← 20 min

```

System 0 is trained specifically for Figure 03. No portability.

The Positioning

System 0 = Apple Silicon — Built for one device, incredibly optimized, proprietary.

SOMA Runtime = ARM — Built for every device, licensable, deployable anywhere.

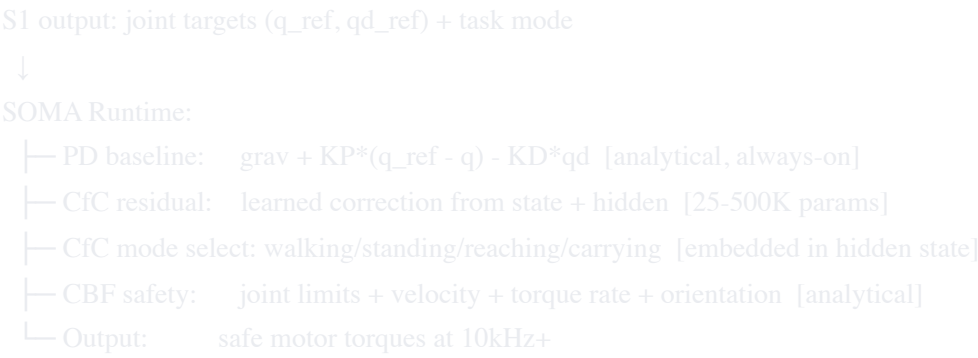
Figure spent \$2.6B building System 0 for one robot. The other 100+ humanoid companies (Unitree, Apptronik, Agility, Collaborative Robotics, Standard Bots, etc.) need their own System 0. They can't afford to build it. They buy SOMA.

7. THE FULL RUNTIME: What SOMA Must Do

Capability Map

Capability	Status	Architecture	Data Needed
Gravity compensation	✓ PROVEN	PD analytical + CfC residual	URDF only
Trajectory tracking	✓ PROVEN (v7)	CfC-NCP via DAgger	Auto-generated expert
Push recovery	✓ PROVEN (v8)	Residual CfC via DiffSim	URDF only
Safety enforcement	✓ PROVEN	CBF analytical	Joint/velocity limits from URDF
Frequency invariance	✓ PROVEN (v7)	CfC dt-gate	Built into training
Walking gaits	✗ NOT PROVEN	CfC + reference trajectory	Option B: optimization-generated
Running/transitions	✗ NOT PROVEN	CfC + curriculum training	Option C: RL teacher
Whole-body coordination	✗ NOT PROVEN	Scaled CfC (200-500K)	Option C+D: RL + mocap
Contact/force control	✗ NOT PROVEN	CfC + F/T sensor input	DiffSim with contact
Impedance modulation	✗ NOT PROVEN (arch natural)	CfC σ -gate = impedance	DiffSim curriculum

Architecture for Full Runtime



Scaling Estimates

Robot	DOF	State Dim	CfC Params	Est. Latency	Est. Training
Franka (done)	7	21	134K	133μs	20 min
Go2 (done)	12	33	26K	65μs	21 min
G1 humanoid	23	55	80-150K	60-80μs	30-45 min
Full humanoid (30+)	30+	75+	200-500K	80-120μs	45-90 min

Even at 500K params, that's still 20x smaller than System 0 and well within 1kHz on CPU.

8. NEXT STEPS (Priority Order)

Immediate (This Week)

- Walking on Go2** — Option B: generate trotting gait reference via simple trajectory optimization, train CfC residual to track it through DiffSim
 - This is THE demo that completes the POC
 - Same custom_vjp infrastructure, longer rollouts (200+ steps)
 - Expected training time: 30-60 min (longer horizon = more FD cost)
- Investor notebook** — Combine v7 arm + v8 quadruped into single demo
 - Hard numbers from both versions
 - Comparison table vs System 0
 - "20 minutes from URDF" narrative

Short-term (2-4 Weeks)

3. **G1 humanoid port** — Same pipeline, different URDF
 - Proves robot-agnostic claim
 - 23 DOF, expect 80-150K params, 30-45 min training
4. **RL teacher pipeline** — Train PPO locomotion teacher, distill into CfC
 - Enables diverse locomotion (turns, speed changes)
 - Uses Unitree's open-source legged_gym as starting point

Medium-term (1-2 Months)

5. **Contact force control** — Add F/T sensor input to CfC state
 - Enables grasping, surface interaction
 - DiffSim with contact models (MJX supports this)
 6. **Whole-body coordination** — Walking + reaching simultaneously
 - Scale CfC to 200-500K params
 - Multi-objective DiffSim cost (balance + tracking + energy)
 7. **Public mocap integration** — AMASS/CMU for natural movement
 - Proves SOMA can produce human-like motion without proprietary data
 - Direct comparison to System 0's motion quality
-

9. SUMMARY: WHAT WE KNOW

The Architecture is Right

CfC + CBF is the correct architecture for a full motor control runtime:

- **CfC dt-gate** handles multiple timescales (1ms reflexes to 1s gait cycles) in one network
- **CfC recurrent state** carries context (gait phase, push history, contact memory) that PD/MPC can't
- **CBF** enforces safety regardless of what CfC outputs
- **Residual** design ensures robustness: PD handles known physics, CfC handles unknown/learned behaviors

The Training Pipeline is Right

DiffSim via custom_vjp is proven:

- Full-fidelity MJX forward (contacts, friction, damping all active)
- Finite-difference backward through control dims (12 dims for Go2)
- 21 min on a single T4 GPU

- Converges from step 1 (residual ensures stability)

The Missing Piece is Locomotion

Everything proven so far is "stand and resist." Walking requires:

- Periodic motor patterns (gait generation)
- Contact scheduling (foot placement)
- CoM trajectory tracking
- Transitions (stand→walk→run→stop)

The CfC architecture supports all of this (σ -gate = oscillation, hidden state = phase). The training method (DiffSim) supports it (long rollouts + contact-aware cost). We just haven't demonstrated it yet.

The Data Strategy is Clear

What	Where It Comes From	Human Effort
Robot model	URDF from manufacturer	Zero
Standing/push recovery	DiffSim auto-generates	Zero
Trajectory tracking	Analytical expert auto-generates	Zero
Walking gait reference	Trajectory optimization auto-generates	Zero
Complex locomotion	RL teacher (PPO) → distill to CfC	Reward design (~1 day)
Natural movement	Public mocap (AMASS/CMU)	Retargeting (~1 week)
Task-specific	Customer demos via teleoperation	Customer provides

Total human data needed: ZERO for core runtime. Public mocap for style. Customer demos for tasks.

This is the fundamental bet: System 0 needs 1,000 hours of mocap + months of tuning per robot. SOMA needs a URDF file and 20 minutes.